THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 3	Each stage triples the count: $3^1 = 3$
2	**(b)** 5	$n + 2 = 5$ (two bases and n rectangles)
3	**(d)** 4	n base vertices + 1 apex $= 4$
4	**(b)** 9	$3n = 9$
5	**(d)** 12	$F + V - E = 2$ gives $E = 6 + 8 - 2 = 12$
6	**(b)** 36	A cube has 12 edges: $12 \times 3 = 36$
7	**(d)** $F + V - E = 2$	Euler's formula
8	**(d)** 9	Each stage triples the count: $3^2 = 9$
9	**(a)** 7	$n + 2 = 7$ (two bases and n rectangles)
10	**(b)** 6	n base vertices + 1 apex $= 6$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Faces $= n + 2 = 5$ (1); vertices $= 2n = 6$, edges $= 3n = 9$ (1).	1+1
12	$F + V - E = 6 + 8 - 12$ (1) $= 2$, so Euler's formula is verified (1).	1+1
13	A cube has 12 equal edges (1); total $= 12 \times 4 = 48$ cm (1).	1+1
14	Faces $= n + 2 = 7$ (1); vertices $= 2n = 10$, edges $= 3n = 15$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Faces $= 4$, vertices $= 4$, edges $= 6$ (2); $F + V - E = 4 + 4 - 6 = 2$ (1).	2+1
16	Stage 2: $3^2 = 9$ (1); stage 3: $3^3 = 27$ (1); more $= 18$ (1).	1+1+1
17	A cuboid has 4 edges of each length (1); total $= 4(5 + 3 + 2) = 40$ cm (1); faces 6, vertices 8 (1).	1+1+1
	OR: Faces $= 5$, vertices $= 5$, edges $= 8$ (2); $F + V - E = 5 + 5 - 8 = 2$ (1).	2+1
18	Stage 3: $3^3 = 27$ (1); stage 4: $3^4 = 81$ (1); more $= 54$ (1).	1+1+1
	OR: A cuboid has 4 edges of each length (1); total $= 4(8 + 4 + 3) = 60$ cm (1); faces 6, vertices 8 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Stage 1: 4 segments of 3 cm (1); stage 2: $4 \times 4 = 16$ segments (1); each of length $= 1$ cm (1); total $= 16 \times 1 = 16$ cm (2).	1+1+1+2
	OR: Stage 1: 3 (1); stage 2: 9 (1); stage 3: 27 (1); each stage multiplies the number by 3 (2).	1+1+1+2
20	Stage 1: 4 segments of 6 cm (1); stage 2: $4 \times 4 = 16$ segments (1); each of length $= 2$ cm (1); total $= 16 \times 2 = 32$ cm (2).	1+1+1+2
	OR: Stage 2: 9 (1); stage 3: 27 (1); stage 4: 81 (1); each stage multiplies the number by 3 (2).	1+1+1+2

Mistakes to avoid:

- Counting faces, edges and vertices of a prism and pyramid with each other's formulas (a pyramid has $2n$ edges, a prism $3n$).
- Forgetting the apex when counting the vertices of a pyramid (a square pyramid has 5, not 4).
- In painted-cube questions, using $n - 2$ for the edge cubes but forgetting to multiply by the number of edges (12) or faces (6).
- Assuming that any six squares joined edge to edge fold into a cube; the net must not overlap when folded.
- Using Euler's formula as $F + V + E = 2$; the correct form is $F + V - E = 2$.
- Counting the stages of a fractal wrongly: the starting figure is stage 0, not stage 1.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 15	$3n = 15$
2	**(c)** 8	$F + V - E = 2$ gives $E = 5 + 5 - 2 = 8$
3	**(c)** 60	A cube has 12 edges: $12 \times 5 = 60$
4	**(d)** 6	Six square faces
5	**(d)** 27	Each stage triples the count: $3^3 = 27$
6	**(b)** 8	$n + 2 = 8$ (two bases and n rectangles)
7	**(a)** 7	n base vertices + 1 apex $= 7$
8	**(b)** 18	$3n = 18$
9	**(a)** 9	$F + V - E = 2$ gives $E = 5 + 6 - 2 = 9$
10	**(a)** 72	A cube has 12 edges: $12 \times 6 = 72$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$F + V - E = 5 + 5 - 8$ (1) $= 2$, so Euler's formula is verified (1).	1+1
12	A cube has 12 equal edges (1); total $= 12 \times 8 = 96$ cm (1).	1+1
13	Faces $= n + 2 = 8$ (1); vertices $= 2n = 12$, edges $= 3n = 18$ (1).	1+1
14	$F + V - E = 4 + 4 - 6$ (1) $= 2$, so Euler's formula is verified (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Faces $= 6$, vertices $= 6$, edges $= 10$ (2); $F + V - E = 6 + 6 - 10 = 2$ (1).	2+1
16	Stage 4: $3^4 = 81$ (1); stage 5: $3^5 = 243$ (1); more $= 162$ (1).	1+1+1
17	A cuboid has 4 edges of each length (1); total $= 4(10 + 6 + 4) = 80$ cm (1); faces 6, vertices 8 (1).	1+1+1
	OR: Faces $= 7$, vertices $= 7$, edges $= 12$ (2); $F + V - E = 7 + 7 - 12 = 2$ (1).	2+1
18	Stage 5: $3^5 = 243$ (1); stage 6: $3^6 = 729$ (1); more $= 486$ (1).	1+1+1
	OR: A cuboid has 4 edges of each length (1); total $= 4(12 + 5 + 3) = 80$ cm (1); faces 6, vertices 8 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Stage 1: 4 segments of 9 cm (1); stage 2: $4 \times 4 = 16$ segments (1); each of length $= 3$ cm (1); total $= 16 \times 3 = 48$ cm (2).	1+1+1+2
	OR: Stage 3: 27 (1); stage 4: 81 (1); stage 5: 243 (1); each stage multiplies the number by 3 (2).	1+1+1+2
20	Stage 1: 4 segments of 12 cm (1); stage 2: $4 \times 4 = 16$ segments (1); each of length $= 4$ cm (1); total $= 16 \times 4 = 64$ cm (2).	1+1+1+2
	OR: Stage 4: 81 (1); stage 5: 243 (1); stage 6: 729 (1); each stage multiplies the number by 3 (2).	1+1+1+2

Mistakes to avoid:

- Counting faces, edges and vertices of a prism and pyramid with each other's formulas (a pyramid has $2n$ edges, a prism $3n$).
- Forgetting the apex when counting the vertices of a pyramid (a square pyramid has 5, not 4).
- In painted-cube questions, using $n - 2$ for the edge cubes but forgetting to multiply by the number of edges (12) or faces (6).
- Assuming that any six squares joined edge to edge fold into a cube; the net must not overlap when folded.
- Using Euler's formula as $F + V + E = 2$; the correct form is $F + V - E = 2$.
- Counting the stages of a fractal wrongly: the starting figure is stage 0, not stage 1.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 8	Eight corners
2	**(c)** 81	Each stage triples the count: $3^4 = 81$
3	**(c)** 9	$n + 2 = 9$ (two bases and n rectangles)
4	**(b)** 8	n base vertices + 1 apex $= 8$
5	**(a)** 24	$3n = 24$
6	**(b)** 12	$F + V - E = 2$ gives $E = 8 + 6 - 2 = 12$
7	**(c)** 84	A cube has 12 edges: $12 \times 7 = 84$
8	**(d)** pyramid	Definition of a pyramid
9	**(b)** 243	Each stage triples the count: $3^5 = 243$
10	**(b)** 10	$n + 2 = 10$ (two bases and n rectangles)

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A cube has 12 equal edges (1); total $= 12 \times 11 = 132$ cm (1).	1+1
12	Faces $= n + 2 = 10$ (1); vertices $= 2n = 16$, edges $= 3n = 24$ (1).	1+1
13	$F + V - E = 8 + 12 - 18$ (1) $= 2$, so Euler's formula is verified (1).	1+1
14	A cube has 12 equal edges (1); total $= 12 \times 15 = 180$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Faces $= 8$, vertices $= 8$, edges $= 14$ (2); $F + V - E = 8 + 8 - 14 = 2$ (1).	2+1
16	Stage 6: $3^6 = 729$ (1); stage 7: $3^7 = 2187$ (1); more $= 1458$ (1).	1+1+1
17	A cuboid has 4 edges of each length (1); total $= 4(9 + 7 + 2) = 72$ cm (1); faces 6, vertices 8 (1).	1+1+1
	OR: Faces $= 9$, vertices $= 9$, edges $= 16$ (2); $F + V - E = 9 + 9 - 16 = 2$ (1).	2+1
18	Stage 7: $3^7 = 2187$ (1); stage 8: $3^8 = 6561$ (1); more $= 4374$ (1).	1+1+1
	OR: A cuboid has 4 edges of each length (1); total $= 4(6 + 6 + 5) = 68$ cm (1); faces 6, vertices 8 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Stage 1: 4 segments of 15 cm (1); stage 2: $4 \times 4 = 16$ segments (1); each of length $= 5$ cm (1); total $= 16 \times 5 = 80$ cm (2).	1+1+1+2
	OR: Stage 5: 243 (1); stage 6: 729 (1); stage 7: 2187 (1); each stage multiplies the number by 3 (2).	1+1+1+2
20	Stage 1: 4 segments of 18 cm (1); stage 2: $4 \times 4 = 16$ segments (1); each of length $= 6$ cm (1); total $= 16 \times 6 = 96$ cm (2).	1+1+1+2
	OR: Stage 6: 729 (1); stage 7: 2187 (1); stage 8: 6561 (1); each stage multiplies the number by 3 (2).	1+1+1+2

Mistakes to avoid:

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